

Task 3: Docker Web Application Deployment using Amazon ECR & EC2

Objective:

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To containerize a web application using Docker, push the image to Amazon ECR, and deploy it on an EC2 instance.

Skills Learned:

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- Linux
- Docker
- Dockerfile
- Docker Images
- Docker Swarm
- Containers
- AWS EC2
- AWS ECR
- Networking
- AWS CLI
- Deployment

Task Requirements

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You must:

- > Create a simple web application using HTML.
- > Containerize it using Docker.
- > Push the Docker image to AWS ECR.
- > Pull and deploy the image with Replica 4 using Docker Swarm.
- > Have 2 Worker Nodes in the Swarm
- > Access the application publicly via EC2 Public IP.

Detailed Instructions

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Phases

Phase 1: Install & configure Docker on EC2

Phase 2: Create web application & Dockerfile

Phase 3: Build & run the image locally

Phase 4: Push image to AWS ECR

Phase 5: Pull & deploy from ECR

Expected output → Application accessible in browser via <http://EC2-Public-IP>